

Toward Robust Assistive Robot Navigation with EEG Motor-Imagery

Mohamed A. Hanafy, Ashutosh Prakash, Saykhun Yusufjonov, Madan M. Rayguru, and Dan O. Popa
Louisville Automation and Robotics Research Institute (LARRI), University of Louisville, Louisville, KY, USA

Overview

Motor imagery is a deliberately generated cue for communicating intent without speech or physical input. In this decision-point shared-control architecture for cooperative robot navigation, the first accepted command was correct at 118 of 120 decision points, and the additional commands were local recovery rather than route failure.

EEG Nonstationarity

Main contribution

- EEG motor imagery can provide a hands-free communication channel.
- EEG is nonstationary: Signal drift across sessions and variability across users limit practical use.
- Under continuous control, one transient decoding error reaches robot motion directly.
- Robust MI EEG decoding, built for inter-session drift and inter-person variability
- Covariance features, **45 s** unlabeled per-session realignment, confidence rejection, temporal confirmation
- Cooperative robot navigation with decision-point shared autonomy

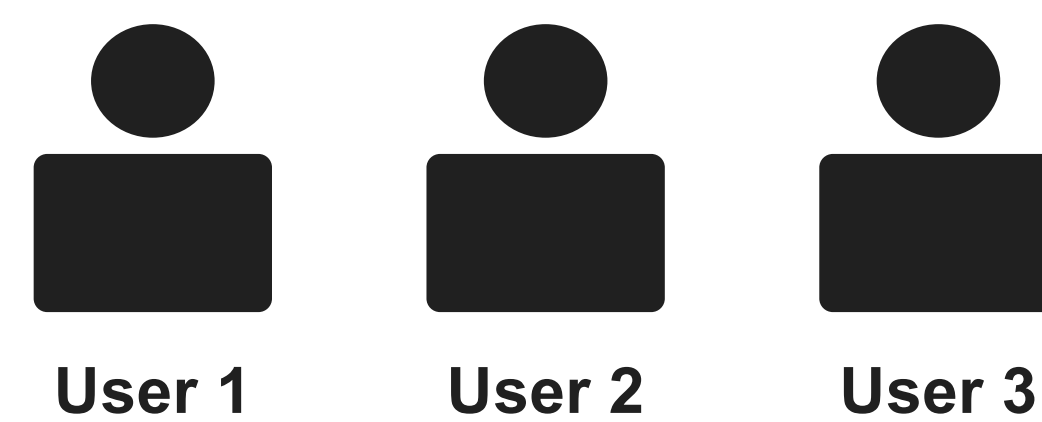


EEG acquisition

15-channel sensorimotor EEG 125 Hz



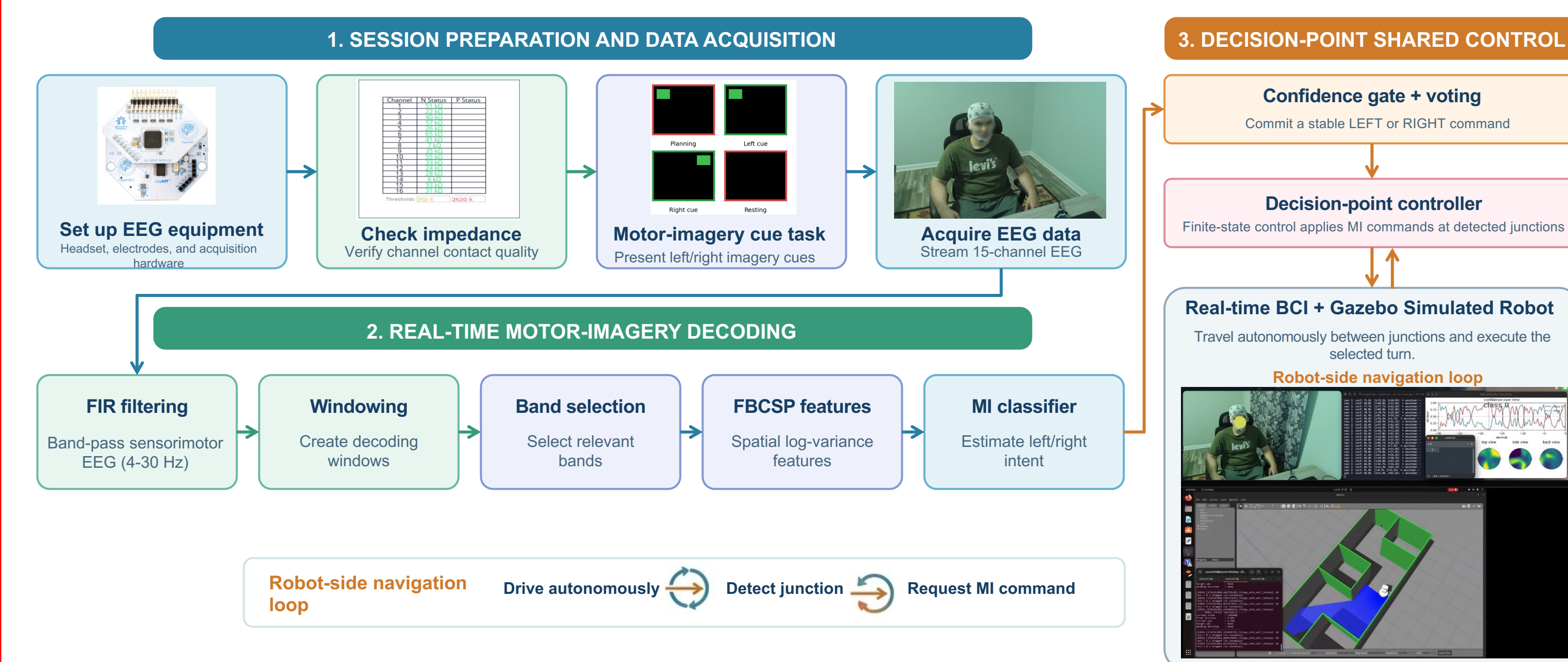
Signals vary across sessions with fatigue, attention, time of day, and physiology.



Inter-person variability
Different users, different patterns

Robust decoding supports reliable, safe human-robot interaction in real-world use.

Closed-Loop Architecture

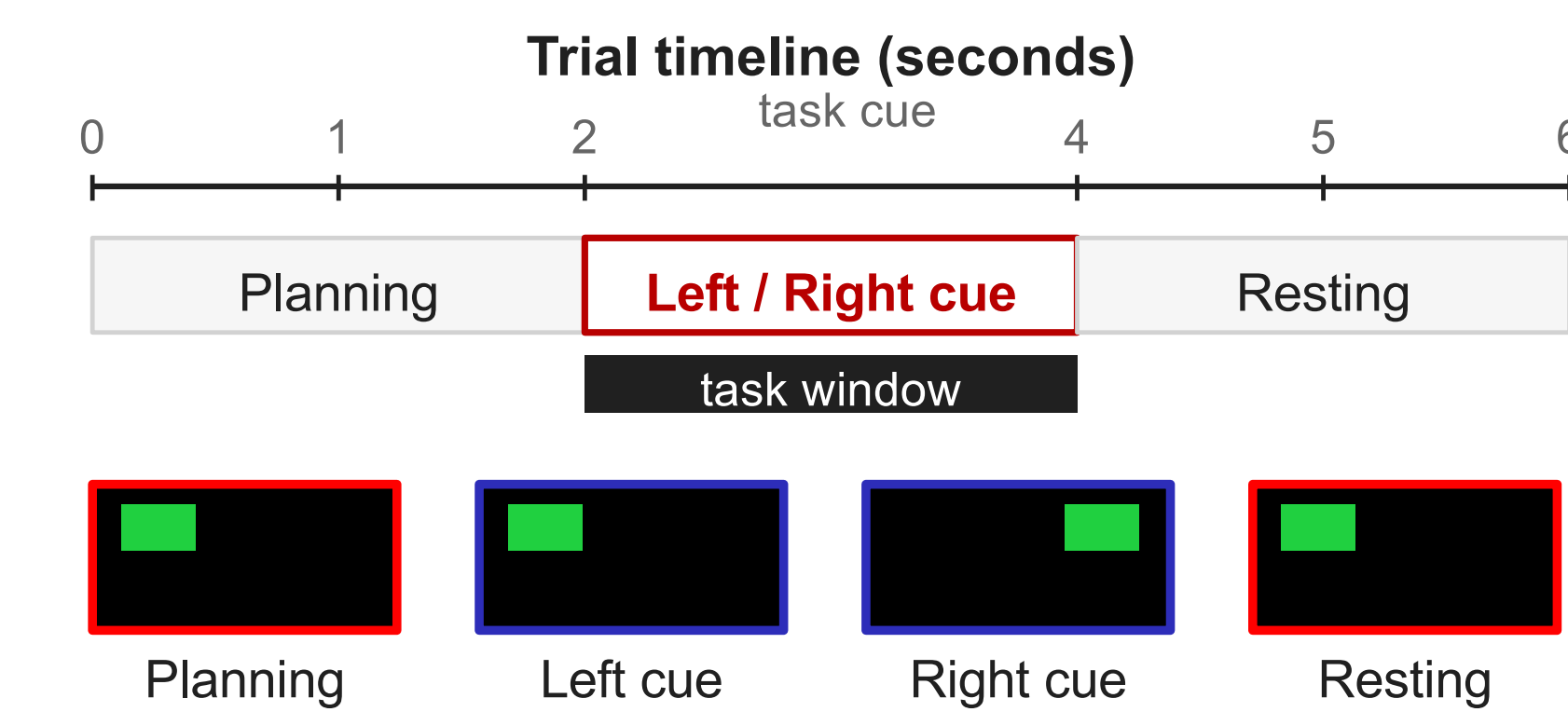


Session preparation, real-time motor-imagery decoding, and decision-point shared control.

An unlabeled calibration block aligns the session and the LDA is recalibrated once.

No labeled navigation data are used after that.

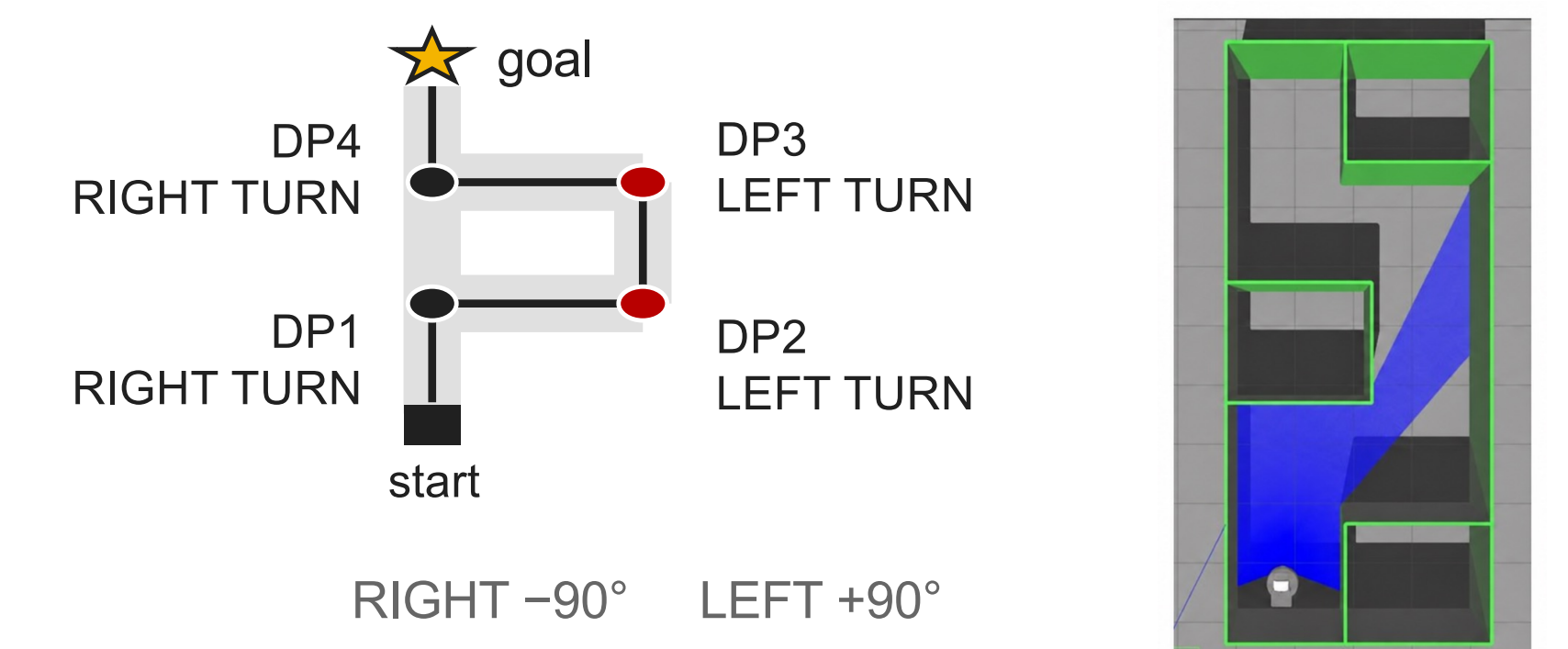
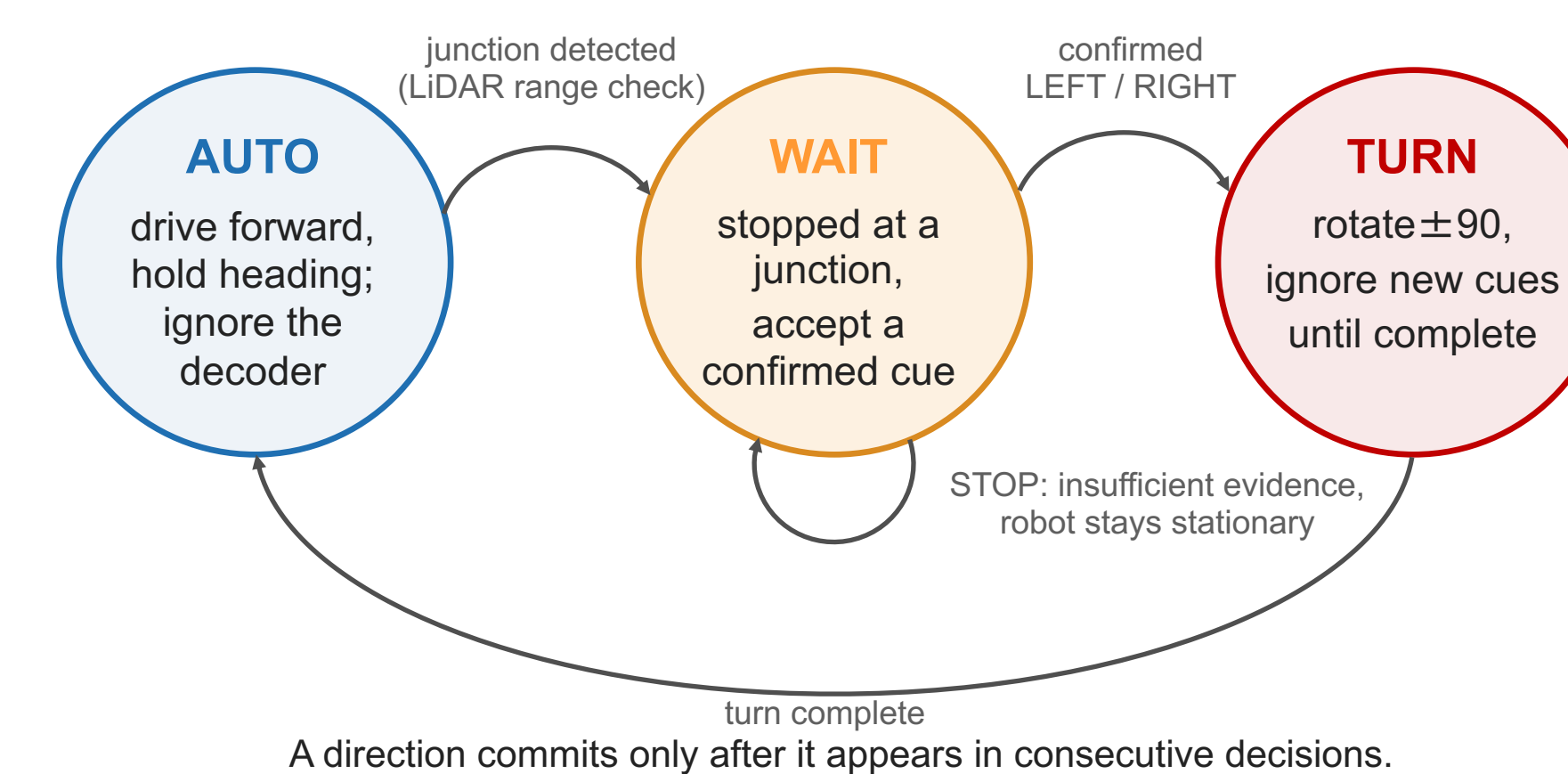
Cue and EEG Decoder



**Binary outputs: LEFT TURN or RIGHT TURN;
Uncertain predictions yield NO DECISION (STOP)**

- 2 s window scored every 0.2 s by a fixed decoder.
- Filter bank 8-12, 11-15, 14-20, 20-30 Hz.
- Covariance features with participant-specific LDA.

Decision-Point Shared Control

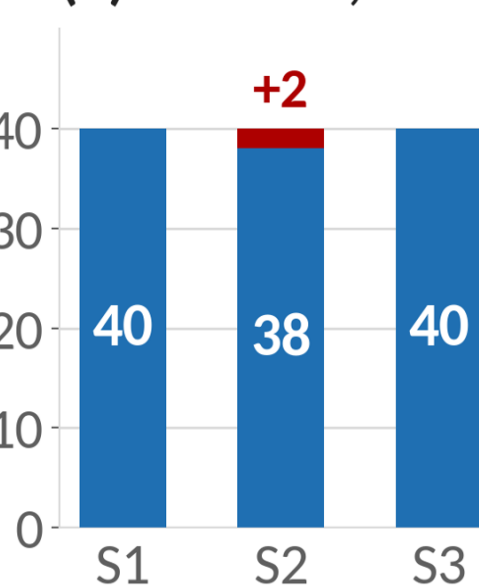


The decoder supplies intent. The robot supplies execution and can override an unsafe action.

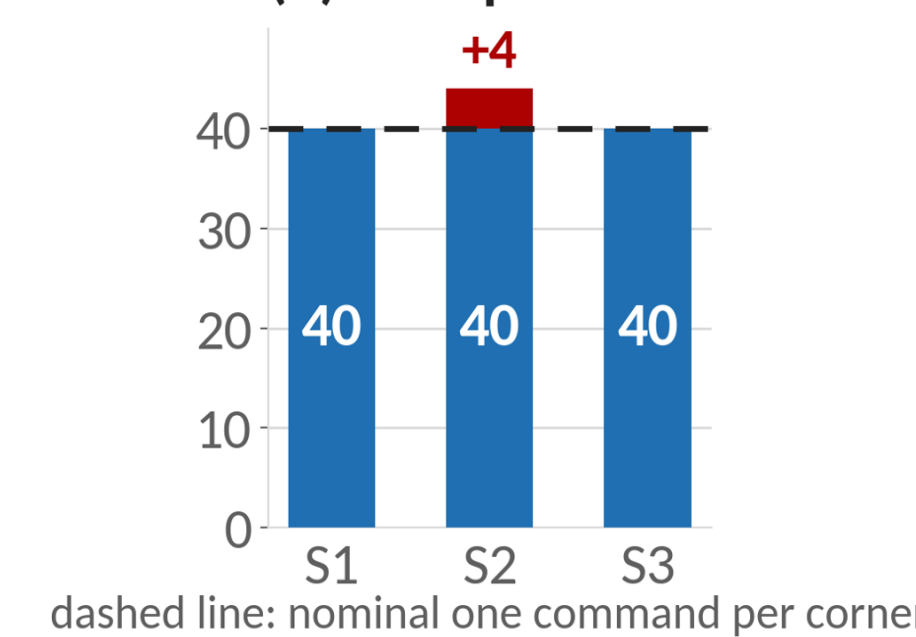
Experiment and Results

3 participants, 10 traversals each, 4 decision points per traversal, 120 total decision points

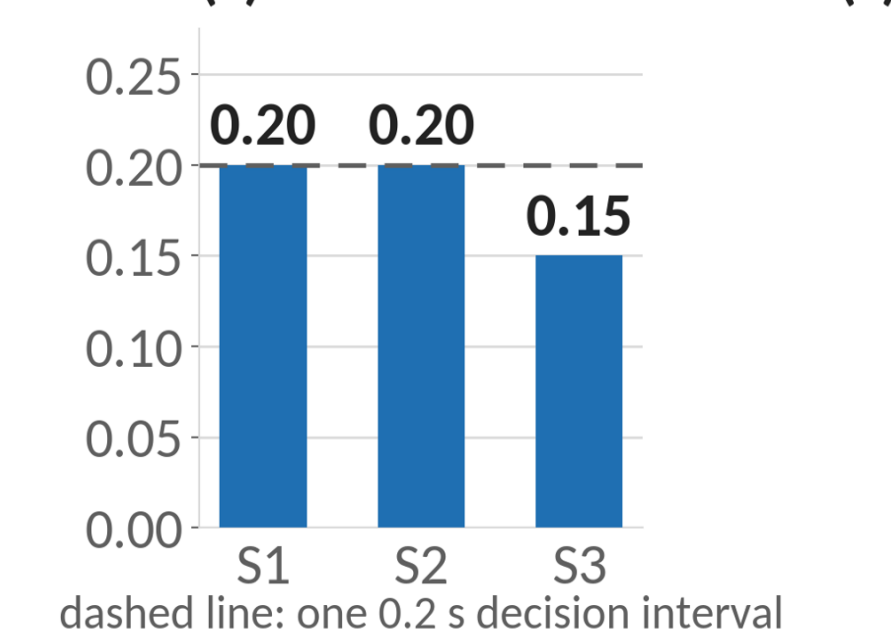
(a) Corners, out of 40



(b) Accepted commands



(c) Median decision wait (s)



98.3%
first accepted command correct

96.8%
successful accepted commands

0.20 s
median decision wait

- All three trials completed. S2 needed four extra commands across two corners; S1 and S3 needed none.
- Task-level outcomes, not offline EEG classification accuracy.

Conclusion and Future Work

Task completion can coexist with an imperfect command sequence when errors are locally recoverable at the point where they occur.

This study uses binary turn decisions; future research can extend the decoder to additional classes. EEG-based control still carries risk, so shared autonomy and safety mechanisms remain essential. All trials ran in simulation with three participants; a physical robot and repeated sessions are the next step.